

Introduction

- **Problem Statement**  
Develop a classification model for planktonic foraminifera using limited labeled 3D volumetric data.
- **Challenge** The dataset is highly imbalanced:
  - ▷ **Labeled:** 210 samples (15 per class across 14 known classes + 1 *unknown* class for noise/fragments)
  - ▷ **Unlabeled:** 18,216 samples
  - ▷ Each sample is a 3D volume of size (128,128,128)

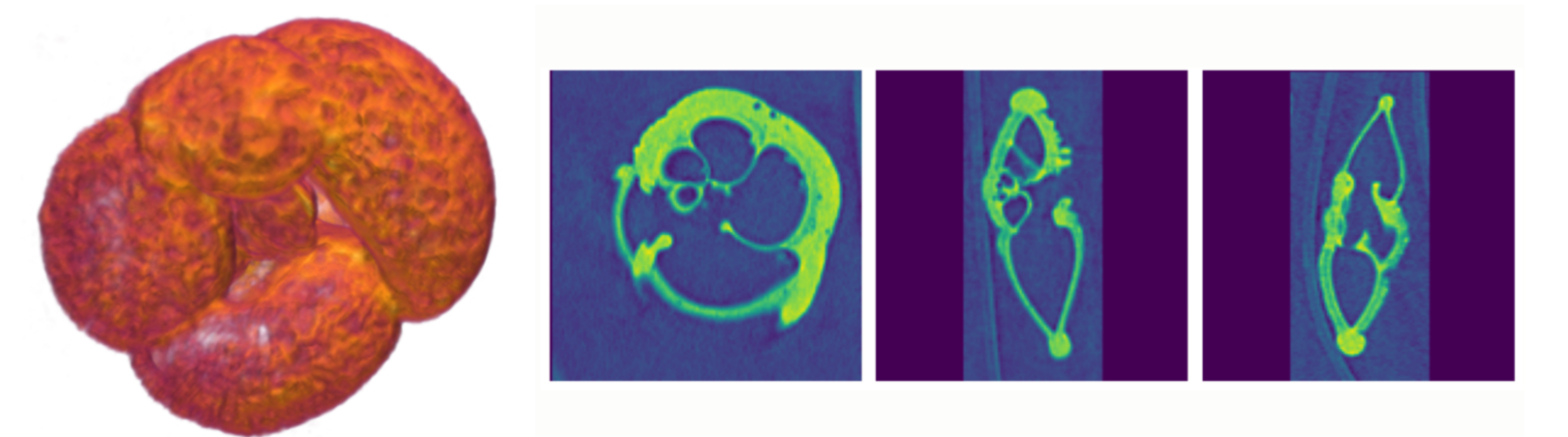


Figure 1: Sample forams from dataset.

Models

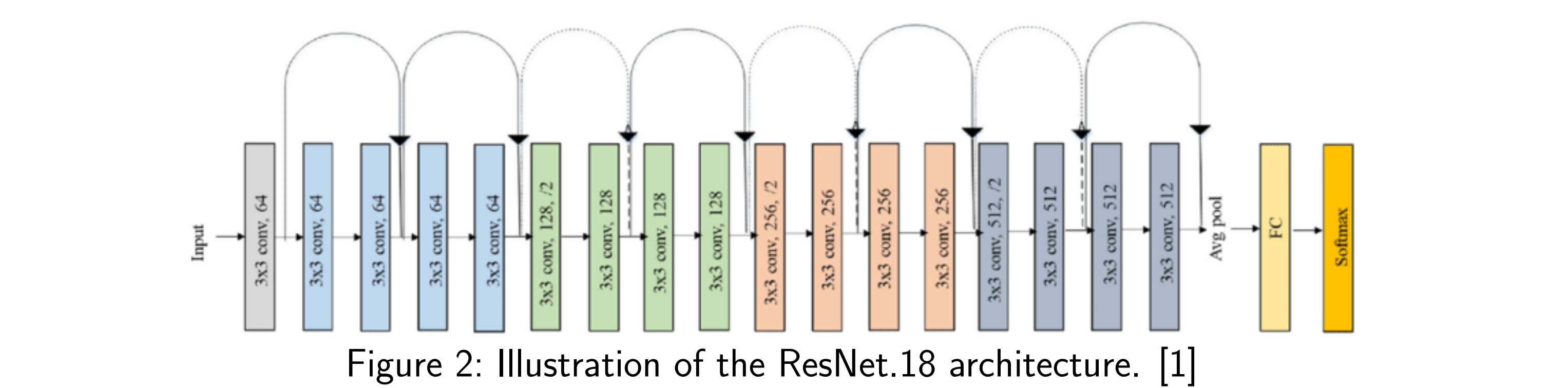


Figure 2: Illustration of the ResNet.18 architecture. [1]

- **ResNet 3D.18**
  - ▷ Trained on complete 3D volumes
  - ▷ **Data Augmentations:** Random 3D Flips, 3D Rotations, Gaussian Noise

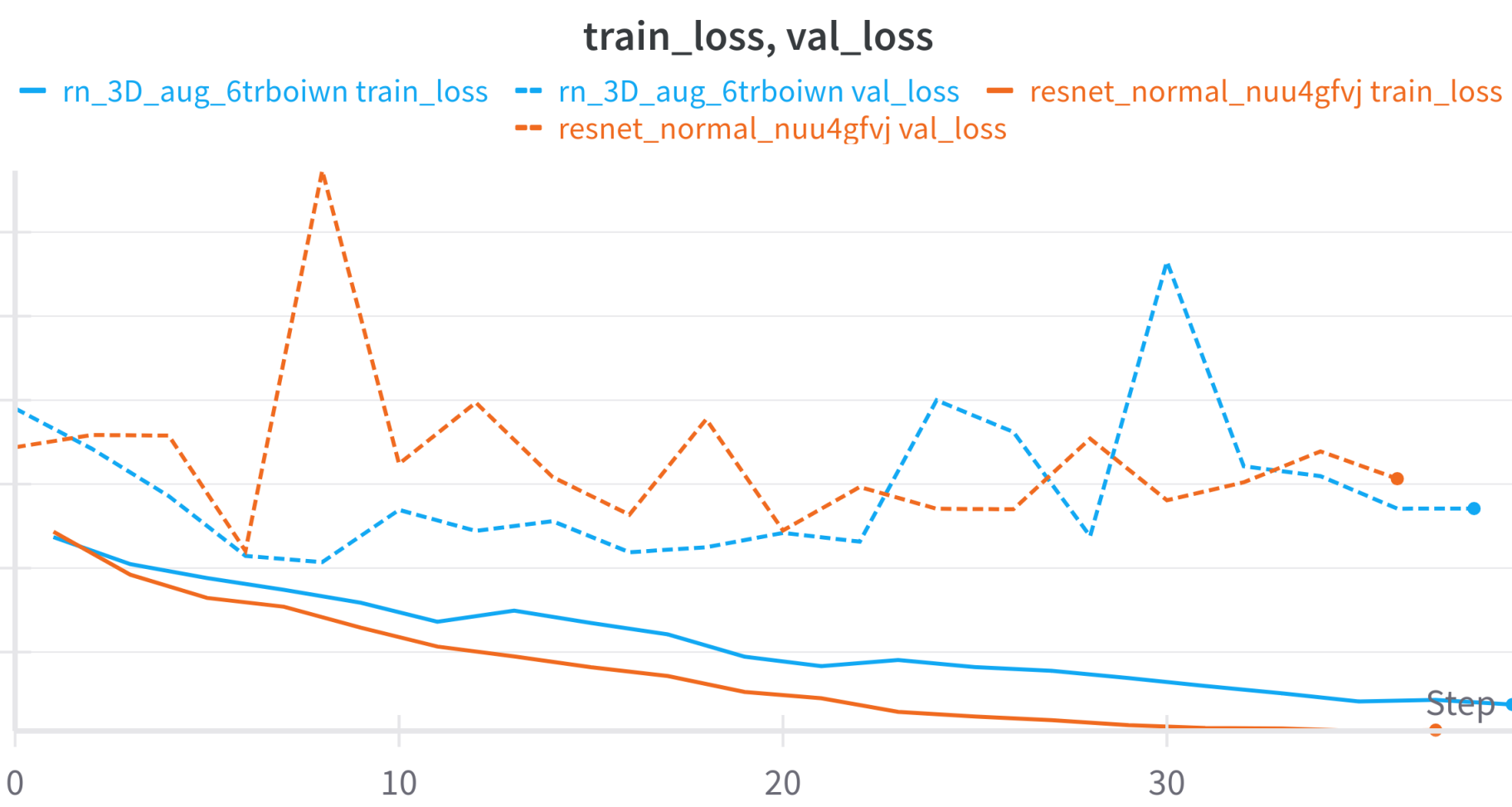


Figure 3: Training and validation losses for the 3D model

- **ResNet 2D**
  - ▷ We used the original ResNet-18 architecture to classify 3D volumes using 2D slice representations
  - ▷ Uses 45 total slices (15 per axis) stacked into a 2D tensor
  - ▷ **Data Augmentations:** H/V Flip, Rotation, Resized Crop, Gaussian Blur

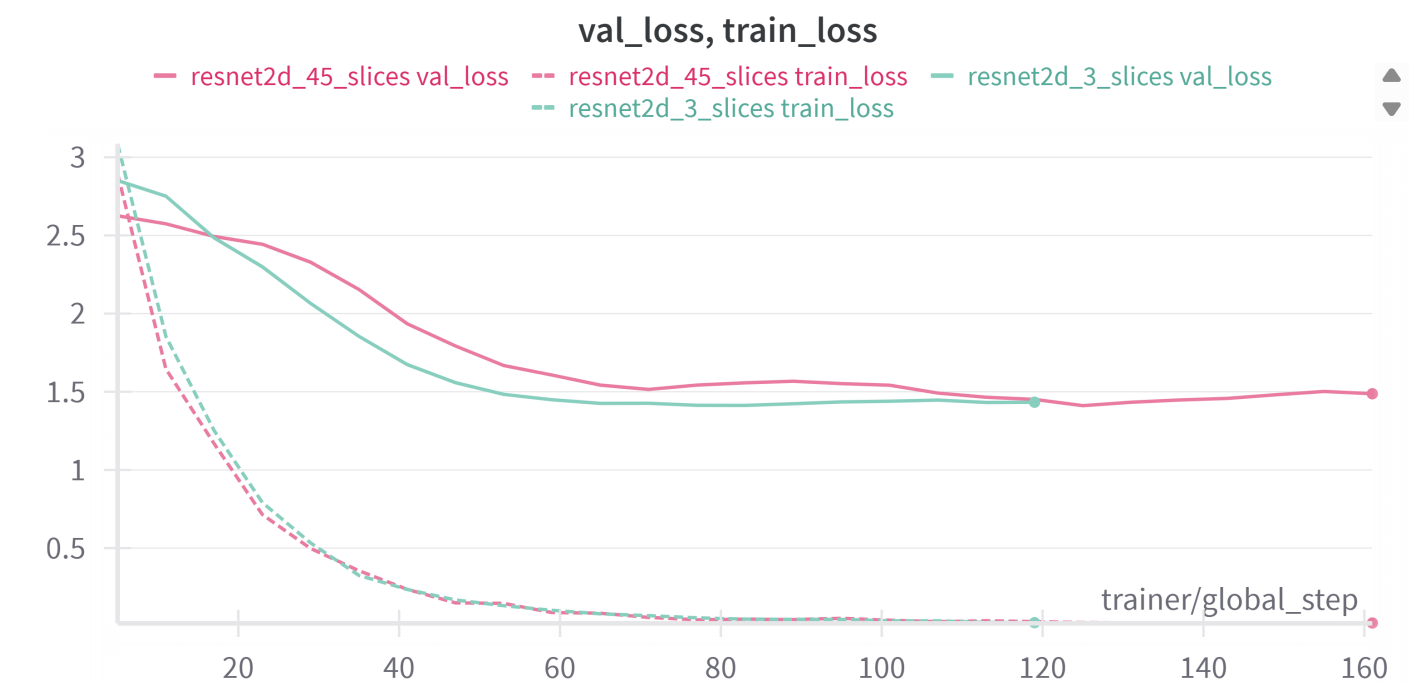


Figure 4: Training and validation losses for the 2D model

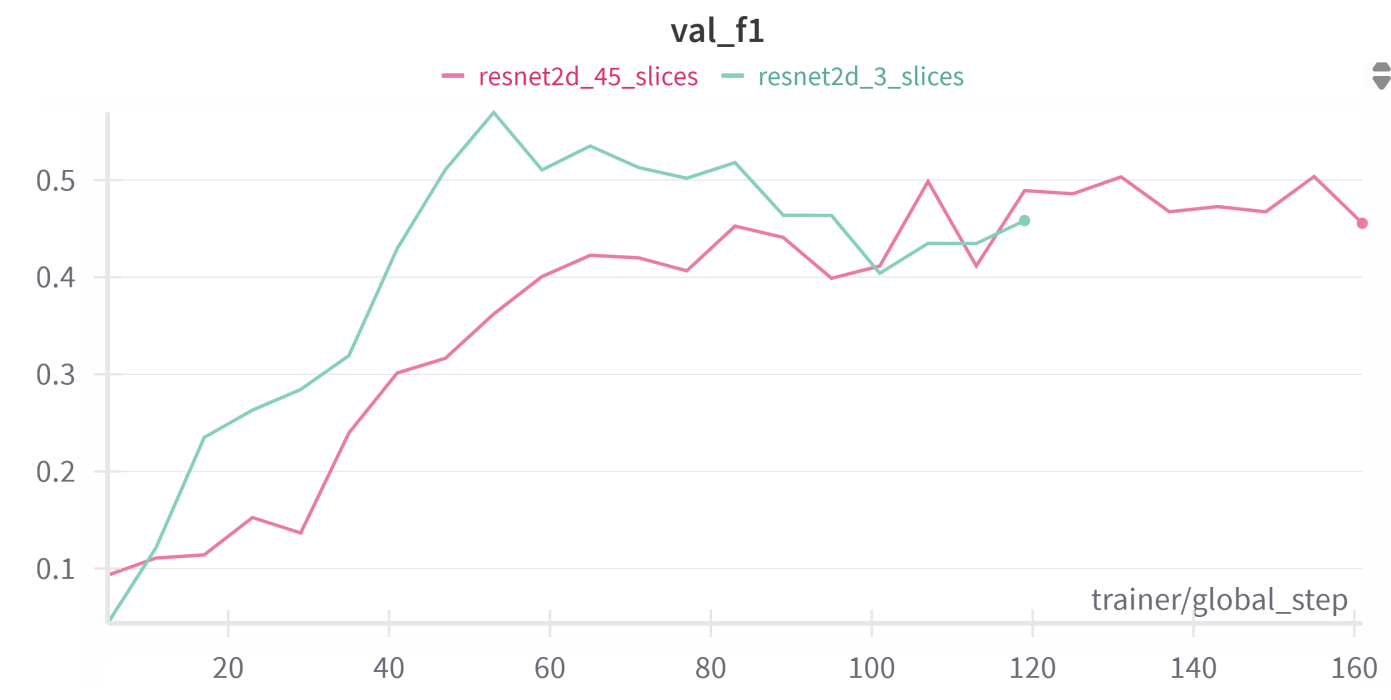


Figure 5: Training and validation F1 scores for the 2D model

Training Parameters:

- **Loss function:** CrossEntropyLoss
- **Optimizer:** Adam
- **Learning Rate:** 1e-4

Performance of Baseline Models on the Test Set:

| Model     | Augmentation Applied | F1 Score |
|-----------|----------------------|----------|
| ResNet 2D | Yes                  | 0.458    |
| ResNet 2D | No                   | 0.4556   |
| ResNet 3D | Yes                  | 0.232    |
| ResNet 3D | No                   | 0.216    |

Self-Supervised Learning

A **Masked Autoencoder (MAE)** was trained on the full dataset to learn meaningful representations of 3D volumetric data. MAE's encoder transforms (128,128,128) volumes into compact (768)-dimensional feature vectors.

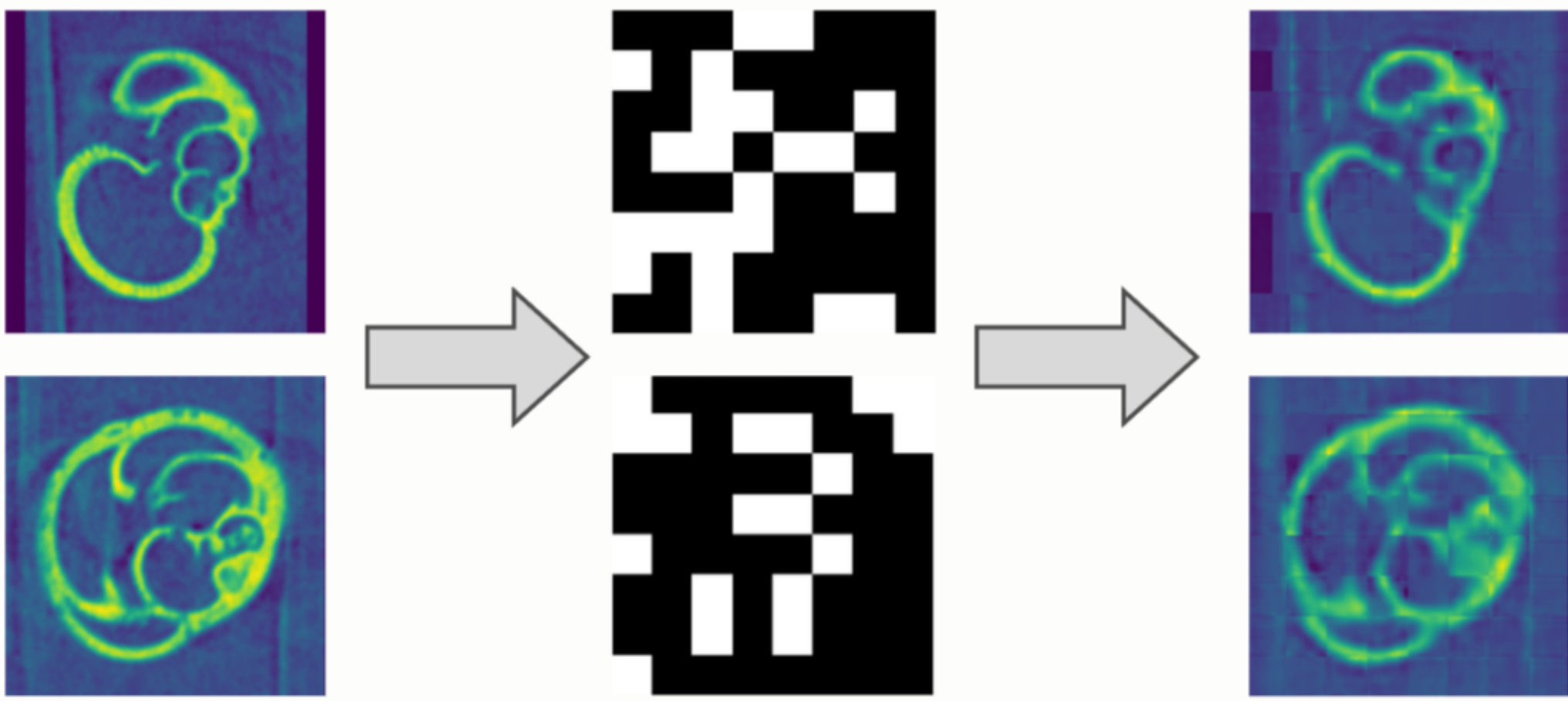


Figure 6: MAE reconstructing masked volume patches.

Semi-Supervised Process:

- Propagate labels to unlabeled samples via feature similarity.
- Retain label predictions with confidence > 0.9.
- Retrain logistic regression on expanded labels.
- Repeat if F1 score improves.

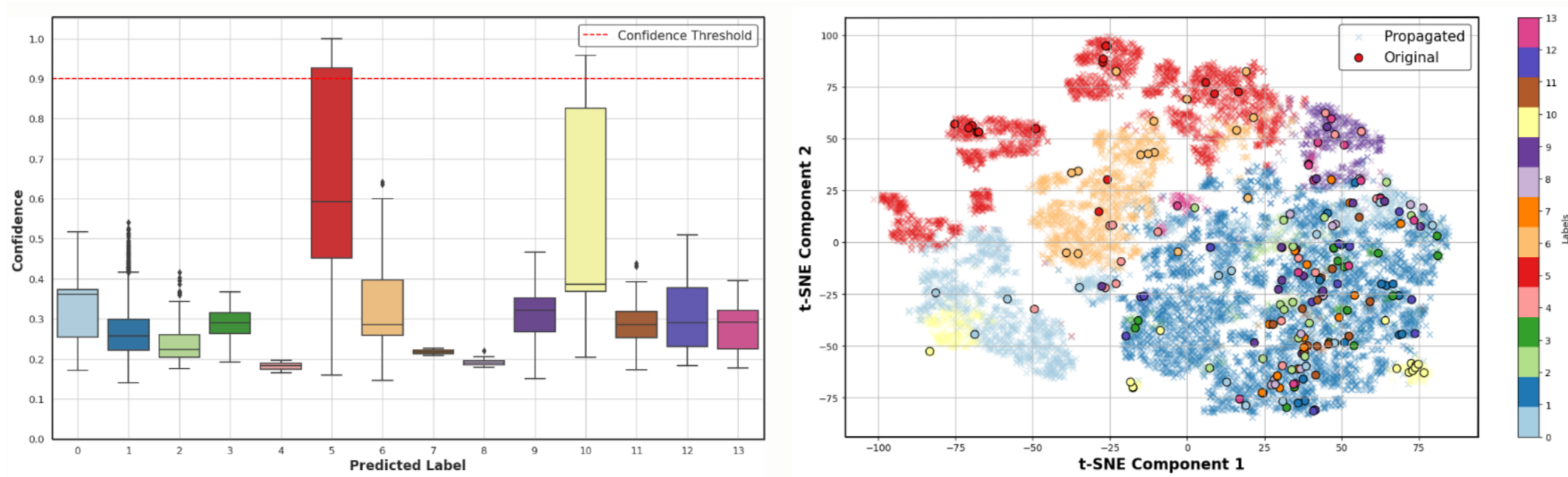


Figure 7: Left: Confidence per label. Right: t-SNE of features with propagated labels.

Performance of Logistic Regression on MAE Features:

| Model              | F1 Score     |
|--------------------|--------------|
| Baseline           | 0.639        |
| With Pseudo-Labels | <b>0.704</b> |

Discussion

- **Baseline Models**
  - ▷ ResNet 3D benefits from 3D-specific augmentations but suffers from unstable validation loss, suggesting overfitting and poor generalization.
  - ▷ ResNet 2D, using central slice representations, achieves smoother convergence and better F1 scores, indicating improved robustness despite reduced input dimensionality.
- **Self-Supervised Learning**
  - ▷ The MAE encoder successfully learns compact and informative representations from 3D volumes.
  - ▷ Two classes form distinct clusters in feature space and are predicted confidently; others remain mixed.

Future Work and Limitations

- Explore alternative 3D architectures like 3D DenseNet. Compared to a ResNet model, it requires more memory but is often more parameter-efficient.
- Apply additional data augmentation techniques, incorporate focal loss tuning or class-balanced sampling targeted to minority classes for handling class imbalance more robustly.
- Adopt alternative self-supervised approaches to learn more discriminative features across all classes.
- Experiment with non-linear classifiers on top of the MAE encoder output, such as a small Multi-Layer Perceptron (MLP).

References

[1] F. Ramzan, M. U. Khan, A. Rehmat, S. Iqbal, T. Saba, A. Rehman, and Z. Mehmood. A deep learning approach for automated diagnosis and multi-class classification of alzheimer's disease stages using resting-state fmri and residual neural networks. *Journal of Medical Systems*, 44, 12 2019.